

Digital Abelian Logical Phase Control in a Correlated-Hopping Ladder

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Abstract

We report an exact-diagonalization study of a digital logical phase primitive in a number-conserving correlated-hopping ladder. The model is a hard-core two-leg Su-Schrieffer-Heeger ladder with a fixed rung-major Jordan-Wigner string convention. A sequential exchange protocol is read out interferometrically in a dressed two-state logical frame, using matched exchange and round-trip controls at opposite statistical angles. We reconstruct the complete logical matrix, including its singular values, leakage and off-diagonal mixing. At $L = 14$ and $N = 2$, all fine-step data at $D = 4, 6, 8, 10, 12$ and $\Delta t = 0.125$ follow $\Delta\phi/\theta = -1 + 0.66493D^{-2.28624}$ with $R^2 = 0.999983$. The retained points have worst-case leakage below 5×10^{-5} and $\sigma_{\min} > 0.999975$. Registered digital path deformations give logical unitary changes below 5.41×10^{-5} at the tested operating point. Repeated cycles remain near the same logical Z axis through $n = 8$. The current gate family commutes. This work establishes an Abelian digital phase primitive for the specified Hamiltonian; it does not demonstrate non-Abelian braiding, universality, fault tolerance or a physical anyon platform.

1 Introduction

Topological quantum computation requires a protected fusion space and physical operations that act noncommutatively on that space [1, 2, 3]. A diagonal logical phase is not sufficient. Nevertheless, a controlled Abelian primitive is a useful intermediate result when its Hamiltonian, encoding, propagation error and claim boundary are all explicit.

Occupation-dependent hopping provides a controlled route to one-dimensional models with tunable statistical phases [4]. In a number-conserving setting, edge and parity structure can be subtle: a number-conserving parent can exhibit Majorana-like properties without being a mean-field Bogoliubov-de Gennes model [5]. These observations motivate the present numerical study, but neither establishes a non-Abelian operation for the ladder studied here.

The objective of this work is narrow. We test whether a frozen, number-conserving correlated-hopping ladder supports a reproducible digital logical phase with a controlled deep-trap limit. The tests are designed to separate physical localization in trap depth D from numerical convergence in the integration step Δt . We also measure path deformations, multi-cycle composition and the commutativity of the resulting gate family.

2 Model and logical protocol

2.1 Correlated-hopping ladder

There are L rungs and two hard-core modes per rung. The rung-major site index is $k = 2j + \sigma$, with rung $j = 0, \dots, L - 1$ and rail $\sigma \in \{0, 1\}$. The fixed Jordan-Wigner convention is

$$a_k = \exp\left(i\theta \sum_{m < k} n_m\right) b_k, \quad n_m = b_m^\dagger b_m. \quad (1)$$

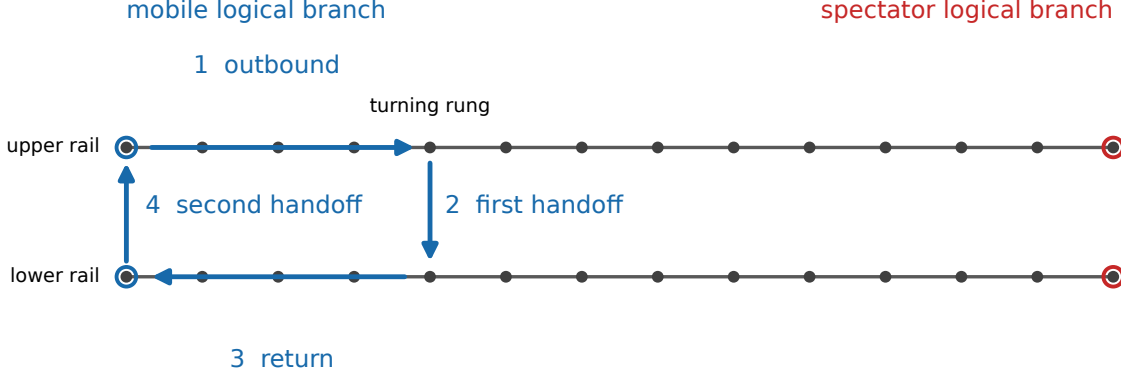


Figure 1: Ordered control sequence for the sequential protocol. The blue arrows show outward transport, the first rail handoff, return transport and the second handoff. The red edge branch is the spectator logical reference. The drawing is a protocol schematic, not a claim of physical particle worldlines in two dimensions.

For $k < l$, the exact-diagonalization Hamiltonian in the fixed- N sector is

$$H(t) = - \sum_{k < l} J_{kl} \left[b_k^\dagger \exp \left(i\theta \sum_{k < m < l} n_m \right) b_l + \text{h.c.} \right] + \sum_k \mu_k(t) n_k. \quad (2)$$

The rung couplings are $J_\perp = 0.1$. Along either rail, the alternating Su-Schrieffer-Heeger couplings are $J_1 = 0.4$ and $J_2 = 1.0$. The statistical angle in Eq. (2) is a property of this fixed correlated-hopping convention. It is not claimed to define convention-independent anyon statistics.

The endpoint Hamiltonian contains deep left and right edge traps. Its two selected low-energy states are rotated into a maximally localized dressed logical frame $\{|0_L\rangle, |1_L\rangle\}$ associated with the left and right edge configurations. This is a logical reference frame for control and readout. It is not asserted to be a topologically protected code.

2.2 Sequential digital exchange

One logical branch is mobile while the other remains trapped as a spectator. The mobile branch is carried outward, transferred by two non-simultaneous handoffs, and returned on the opposite rail. A matched round trip omits the rail exchange. The two protocols have identical endpoint Hamiltonians and are compared at $+\theta$ and $-\theta$. Figure 1 shows the sequence schematically.

For an isolated handoff, the local two-state Hamiltonian may be written as

$$H_\varphi(t) = \begin{pmatrix} \epsilon_1(t) & -J \exp(i\varphi) \\ -J \exp(-i\varphi) & \epsilon_2(t) \end{pmatrix}. \quad (3)$$

It obeys $H_\varphi = G_\varphi H_0 G_\varphi^\dagger$ with $G_\varphi = \text{diag}(\exp(i\varphi), 1)$. If the avoided crossing remains open and the handoff is adiabatic, the source state is transferred to the target with an oriented factor $\exp(-i\varphi)$ up to a common phase. In the strictly localized limit of the implemented sequence, one longitudinal transfer crosses the occupied intermediate mode and contributes $\Delta\phi = -\theta$; the matched round trip cancels its forward and reverse contributions.

At finite trap depth, virtual configurations outside the localized handoff subspace correct the phase, amplitude and leakage. The numerical test is thus

$$\frac{\Delta\phi}{\theta} = -1 + aD^{-p} + O(D^{-p-1}), \quad (4)$$

with the adiabatic duration scaled as $T(D) = T_{\text{ref}}(D/D_{\text{ref}})^2$.

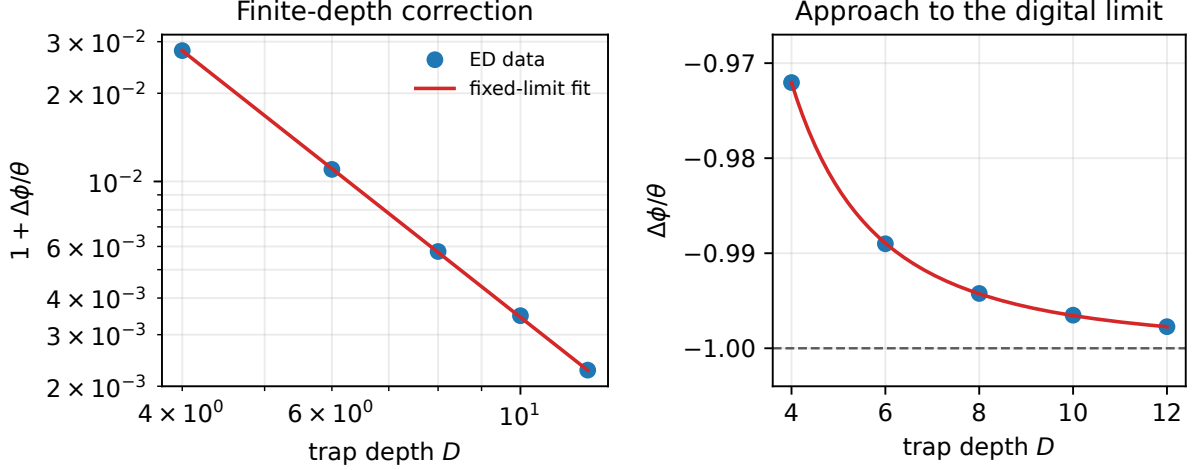


Figure 2: All-fine-step deep-limit data. Left: the correction $1 + \Delta\phi/\theta$ on logarithmic axes. Right: the raw slope and the fixed limiting value -1 . The curve is a two-parameter fit, not an independently derived exponent.

3 Numerical method and diagnostics

All results use exact diagonalization at $L = 14$ and $N = 2$. The digital propagator uses a symmetric split step: half of the diagonal trap potential, the exactly exponentiated hopping matrix, then the second potential half step. For each signed angle and for exchange and round-trip protocols, the projected logical map is

$$S = U_0^\dagger U(T) U_0, \quad (5)$$

where U_0 is the endpoint logical frame. Its polar unitary is used for logical phases, while the singular values of S retain loss information.

Even-in- θ contributions are removed by combining the matched exchange and round-trip unitaries at opposite angles. The resulting odd map is diagonalized only after the full 2×2 logical reconstruction. The reported diagnostics are the odd phase slope $\Delta\phi/\theta$, worst-case leakage $1 - \sigma_{\min}^2$, the minimum singular value, off-diagonal norm, and the overlap-tracked isolation gap between the two selected logical branches and the remaining spectrum during the handoffs.

The deep-limit campaign uses $\theta = 0.3$, $D = 4, 6, 8, 10, 12$, $T_{\text{ref}} = 20000$, $D_{\text{ref}} = 4$ and $\Delta t = 0.125$. Independent time-step data at $D = 6$ and $D = 8$ use $\Delta t = 0.5, 0.25, 0.125$. The finite-depth path campaign uses $D = 8$, $T = 80000$ and $\Delta t = 0.25$. The composition campaign uses $D = 6$, $T = 45000$, $\Delta t = 0.25$ and $n = 1, 2, 4, 8$ cycles.

4 Results

4.1 Deep-trap and time-step closure

The five all-fine-step depth points are given in Table 1. A fit to Eq. (4) with the limiting value fixed to -1 gives

$$a = 0.66493, \quad p = 2.28624 \pm 0.00764, \quad R^2 = 0.999983. \quad (6)$$

The residual root-mean-square error is 3.91×10^{-5} . All five retained points have leakage below 5×10^{-5} and a positive sampled handoff isolation gap. Figure 2 displays both the finite-depth correction and the approach to the fixed digital limit.

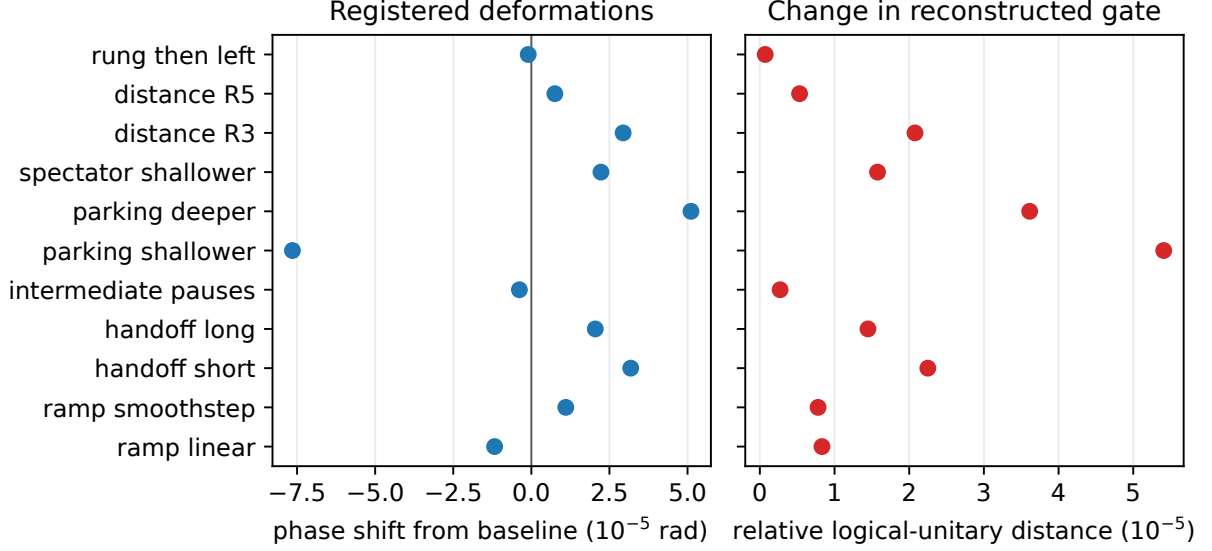


Figure 3: Effect of the ten non-baseline protocol deformations. Left: odd phase shift relative to the baseline. Right: distance between reconstructed logical unitaries after removal of the common global phase.

Table 1: Fine-step deep-limit data at $\theta = 0.3$ and $\Delta t = 0.125$. The protocol duration follows $T(D) = 20000(D/4)^2$.

D	T	$\Delta\phi/\theta$	leakage	$\sigma_{\min}$	handoff gap
4	20000	-0.972041	3.02e-05	0.999985	6.68e-04
6	45000	-0.989005	1.14e-05	0.999994	5.30e-03
8	80000	-0.994233	2.25e-05	0.999989	4.43e-03
10	125000	-0.996517	3.42e-05	0.999983	1.27e-04
12	180000	-0.997730	4.97e-05	0.999975	8.76e-03

The time-step study is reported in Table 2. At both $D = 6$ and $D = 8$, the coarse $\Delta t = 0.5$ rows exhibit substantial leakage and are explicit numerical failures. Reducing to $\Delta t = 0.25$ restores the intended gate, while the $\Delta t = 0.125$ rows have leakage near 10^{-5} . Thus the depth trend used in Fig. 2 is not inferred from the coarse integrator.

4.2 Protocol deformations and composition

Eleven registered deformations preserve the sequential protocol and endpoint Hamiltonian while varying ramp profile, handoff duration, pauses, trap depths, separation or handoff order. The largest phase shift from the baseline is 7.65×10^{-5} radians and the largest relative logical-unitary distance is 5.41×10^{-5} at the tested operating point (Table 4 and Fig. 3). This is finite-depth protocol robustness, not a standalone proof of topological protection.

For repeated cycles, the phase additivity error reaches 2.57×10^{-2} radians at $n = 8$. This residual is reported rather than absorbed in phase unwrapping. The worst-case leakage remains between 2.12×10^{-4} and 2.46×10^{-4} over the four tested values of n , and the rotation axis remains close to Z (Table 3 and Fig. 4). There is no observed superlinear leakage signature within this four-point test, but this does not establish a long-depth fault model.

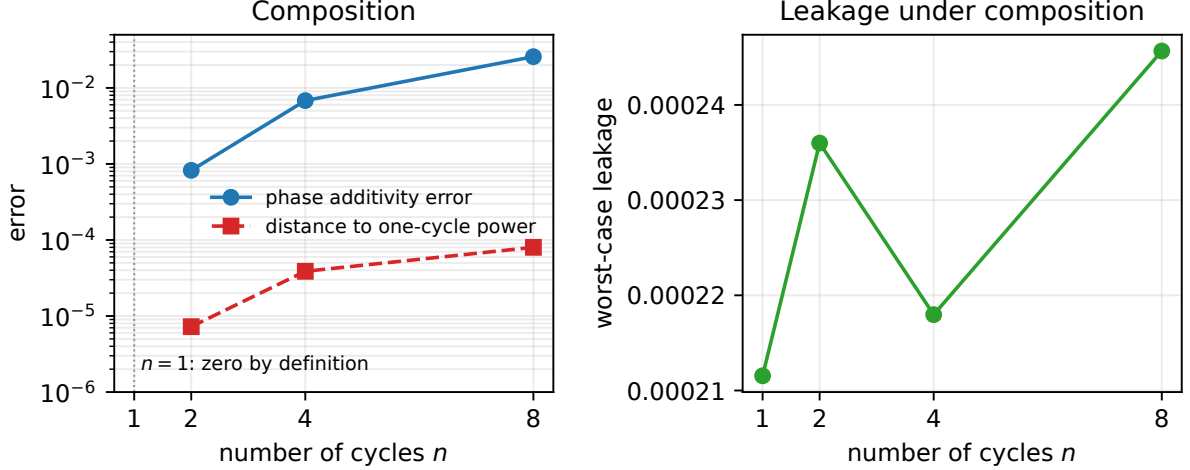


Figure 4: Repeated-cycle composition. The one-cycle phase-additivity error is identically zero and is not placed on the logarithmic axis. Leakage remains small over the tested range, while the finite coherent phase error grows with cycle count.

Table 2: Independent time-step check. The $\Delta t = 0.5$ rows are retained as a resolved coarse-step failure, rather than being used in the closure fit.

D	Δt	$\Delta\phi/\theta$	leakage	$\sigma_{\min}$
6	0.500	-0.926576	4.20e-01	0.761595
6	0.250	-0.990352	2.12e-04	0.999894
6	0.125	-0.989005	1.14e-05	0.999994
8	0.500	3.392471	9.99e-01	0.028334
8	0.250	-0.995348	4.71e-04	0.999765
8	0.125	-0.994233	2.25e-05	0.999989

4.3 Commutativity

The reconstructed gates are effectively diagonal Z rotations. An earlier three-angle audit gave commutator norms of 4.49×10^{-7} , 5.32×10^{-7} and 2.89×10^{-7} for angle pairs $(0.3, 0.6)$, $(0.3, 0.9)$ and $(0.6, 0.9)$, respectively. These values are numerical residuals around the expected commuting family, not evidence of a second logical generator. The present operation therefore cannot by itself realize a non-Abelian braid or a universal gate set.

5 Discussion and claim boundary

The main result is a reproducible digital Abelian phase primitive for the Hamiltonian in Eq. (2). The finite-depth sequence agrees with the localized two-level transfer rule, and the independent time-step study separates physical trap-depth convergence from a coarse-step artifact. The full logical reconstruction is important here: phase alone would not detect the failures at $\Delta t = 0.5$ or quantify the retained leakage.

Several boundaries are essential. First, the Jordan-Wigner string is fixed by the rung-major representation and the result is tied to that correlated-hopping model. Second, the calculation is an exact-diagonalization result at $L = 14$, $N = 2$; it is not a thermodynamic-limit or experimental robustness certificate. Third, finite path deformations and near-diagonal composition do not make the gate topological. Finally, a family satisfying

$$[U_Z(\theta_1), U_Z(\theta_2)] = 0 \quad (7)$$

Table 3: Repeated-cycle composition at $D = 6$, $\theta = 0.3$ and $\Delta t = 0.25$.

n	phase error (rad)	leakage	one-cycle-power distance	Z -axis drift
1	0.000e+00	2.12e-04	3.14e-16	1.50e-07
2	8.232e-04	2.36e-04	7.24e-06	8.01e-08
4	6.809e-03	2.18e-04	3.87e-05	1.97e-07
8	2.574e-02	2.46e-04	8.01e-05	2.38e-08

Table 4: Summary of the eleven finite-depth protocol deformations at $D = 8$, $\theta = 0.3$ and $\Delta t = 0.25$.

quantity	maximum value
largest phase shift	7.648e-05
largest logical-unitary distance	5.408e-05
largest error to $-\theta$	1.447e-03

cannot supply the noncommuting exchanges required for a non-Abelian braid.

The next research question is therefore not whether to relabel this operation as a braid. It is whether a microscopic extension can supply a locally protected fusion space and two derived exchanges with a nonzero commutator. That question requires separate evidence for locality, a protected code space, physical control terms and a non-Abelian algebra.

6 Conclusion

The ANTLEER ladder supports a numerically closed Abelian digital logical phase primitive in the registered $L = 14$, $N = 2$ protocol. The fine-step depth scan approaches $\Delta\phi = -\theta$ with a quantified finite-depth correction, and the associated controls bound leakage, numerical resolution, finite-depth deformations and short repeated-cycle behavior. The results do not establish non-Abelian statistics or topological quantum computation. They provide a reproducible baseline against which any future non-Abelian extension must be tested.

Data and code availability

All scripts, numerical JSON files, figure-generation code and the logical closure check are available at github.com/infinition/antler. The public repository includes the Phase 0 to 4.7 reproducibility archive.

References

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